

Cybersecurity and Ethical Hacking Fundamentals

Core Principles of Cybersecurity

Cybersecurity is the practice of protecting systems, networks, programs, and data from digital attacks, unauthorized access, damage, or theft. Ethical hacking (penetration testing or white-hat hacking) is a proactive, legal approach where authorized professionals simulate attacks to identify and fix vulnerabilities.

Key Principles:

- **Risk Management:** Identifying, assessing, and mitigating threats.
- **Defense in Depth:** Multiple layers of security.
- **Least Privilege:** Minimum access rights needed.
- **Zero Trust:** Verify everything, trust nothing.
- **Continuous Monitoring & Incident Response.**

The CIA Triad

The foundational model for information security:

- **Confidentiality:** Information accessible only to authorized users. (Encryption, access controls)
- **Integrity:** Accuracy and trustworthiness of data. (Checksums, digital signatures)
- **Availability:** Systems and data accessible when needed. (Redundancy, backups)

Types of Hackers

Type	Description	Legality & Intent
White Hat	Ethical hackers with permission	Legal, beneficial
Black Hat	Malicious hackers for gain or harm	Illegal
Grey Hat	Hack without permission but report findings	Usually illegal
Script Kiddies	Inexperienced users of ready-made tools	Often illegal
Hactivists	Political or ideological motives	Usually illegal
State-Sponsored	Government-backed groups	Illegal in targets
Insiders	Employees misusing access	Varies

Common Threats

1. Malware

- **Types:** Viruses, Worms, Trojans, Spyware, Rootkits.
- **Impact:** Steals data (Confidentiality), corrupts files (Integrity), slows systems (Availability).

2. Phishing

- Social engineering via fake emails/websites.
- Impact: Credential theft, entry point for other attacks.

3. Ransomware

- Encrypts files and demands ransom.
- Impact: Major Availability loss, financial damage.

Other threats: DDoS, SQL Injection, Zero-Day exploits, Supply Chain attacks.

Defense Recommendations

- Strong passwords + MFA
- Regular patching and updates
- Security awareness training
- Backups (3-2-1 rule)
- Antivirus/EDR tools
- Ethical hacking exercises

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